



2.7.6 Wrap-Up: Reflection

1. How “in the zone” do you feel right now about multiplying polynomials?



Unit 2, Lesson 8: Adding, Subtracting, and Multiplying Polynomials Using More Than One Operation



Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.



Learning Targets

- ☐ I can recognize adding, subtracting, or multiplying a polynomial will result in a polynomial.
- ☐ I can add, subtract, and multiply polynomial expressions using more than one operation.



2.8.1 Warm-Up: Electric Vehicle Technician

Ben Grandy went from studying landscape architecture to becoming an electric vehicle technician because of his passion for fighting climate change.

Ben shared that he believes communication is a very important skill, regardless of your job title in a company. Asking questions to have clarity on product needs, being able to improve products, and always striving to achieve better helps you to grow.

1. Describe a time when you sought to improve in something.
2. Describe a time when clear communication helped you improve.



2.8.2 Collaboration: Adding and Subtracting Polynomials With Distribution

When performing operations on polynomials, the order of operations applies.

Work with your partner to complete the following.

1. The order of operations can be applied to simplify the polynomial $8x + 3(11x - 2)$.
 - a. Rewrite the expression in each step using the description.

	$8x + 3(11x - 2)$	Description
Step 1		Distribute 3 to $(11x - 2)$.
Step 2		Combine like terms.

- b. Complete the statement.

☐ Addition and subtraction

☐ Multiplication

 is done before

☐ addition and subtraction

☐ multiplication

 in the order of operations.

2. Use the order of operations to simplify the polynomial and describe each step you perform.

$10\left(\frac{2}{5}m - 3p^2\right) + 2m - (7p^2 + 6)$	Description

3. Complete the statement.

When adding and subtracting polynomials that include

distribution, first

- distribute monomials,
- add or subtract like terms,

then

- distribute monomials.
- add or subtract like terms.



2.8.3 Activity: Polynomial Scavenger Hunt

- With your partner, complete the polynomial scavenger hunt.
 - Start at one of the Polynomial Cards posted.
 - Draw (or describe) your card's emoji and write the polynomial expression on the recording sheet.
 - Simplify the expression by applying the order of operations.
 - Use your simplified answer to find your next card posted somewhere in the room.
 - Record the emoji on that Polynomial Card and your next expression to simplify.
 - Repeat until you have completed 5 problems.

Polynomial Emoji Scavenger Hunt Recording Sheet	
Card Emoji	Polynomial Expression
	Answer:
	Answer:
	Answer:
	Answer:
	Answer:



The types of expressions you and your partner simplified in the emoji scavenger hunt were all polynomial expressions.

2. Review your answers. Discuss with your partner what the results have in common.

3. Complete the statement.

The result of adding, subtracting, and/or multiplying polynomials will
☐ always
☐ not always
 result in a polynomial.



2.8.4 Wrap-Up: Growth Mindset

1. Check the description that matches your answer to each question for today's lesson.

Mindset Question	Best Description for Me Today
Did I use the partner discussions as learning opportunities?	○ Never ○ Sometimes ○ Usually ○ Always
Did I work on problems that challenged me?	○ Never ○ Sometimes ○ Usually ○ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	○ Never ○ Sometimes ○ Usually ○ Always
Did I put as much effort as I possibly could into the problems?	○ Never ○ Sometimes ○ Usually ○ Always

Unit 2, Lesson 9: Dividing Polynomials



Benchmark

MA.912.AR.1.4 Divide a polynomial expression by a monomial expression with rational number coefficients.



Learning Target

- ☐ I can divide a polynomial expression by a monomial expression.